

Taking down SI and Liver and embedding in OCT

Project: Spatial_Perturb

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WEDNESDAY, 3/25/2026

- To grab before going to the mouse room:
 - 10 cm plates for dissecting the gut in
 - PBS
 - Fixative BD cytofix cytoperm 1:2 dilution into PBS for a total of 15*5ml.
 - Syringe for pumping out the fixative.
 - The tips for the end of the syringe.
 - The labeled sticks to push through the guts.
 - Eye dropper to wet the labeled stick.
 - Plastic box for holding the rods after guts are on them. Fill with 4% PFA
 - 15 - 15 ml conicals for liver. Fill these with PFA **WHAT PERCENTAGE PFA**
- Put PBS in a 10 cm dish.
- Get plastic box for rods ready.
- Cut open mouse's skin - don't nick the inside.
- Then open the inside layer of skin
- cut the lowest possible point, where the colon attaches to the butt
- cut the point that connects to stomach.
- Put the whole gut into the 10 cm dish
- Take Liver, cut the different lobes apart.
 - Put 2 lobes in the flow plate.
 - Put the rest of the lobes in a PFA conical.
- Take spleen and put in flow plate.
- Back to gut.
- cut off the colon from the rest
- Grab the syringe and put tip on it. pull up the 2% fixative - 5 ml.
- Hold top of gut with forceps while flushing it with the syringe.
- Cut off some of the connective tissue after flushing.
- Grab one of the long labeled rods. Wet it with PBS with the eye dropper.
- Grab the top of the gut again, slide it onto the rod.
- Cut off the rest of the gut that's not on the rod.
- Put the rod in the plastic box.
- Wrap up everything downstairs, and go upstairs to the fume hood.
- To grab once you get upstairs:
 - Grab one of the really big plates, put it face down.

- Grab the good pair of scissors with the ballpoint.
- Grab a little beaker, fill it with PBS.
- Grab the box with the elongated rows for the guts. Fill it with 4% PFA. Label it 1 to whatever with a piece of tape.
- Grab the pink base needles with the wax on the end for rolling.
- Grab the 30 gage pins
- 6wp with 4% PFA in each well. Labeled for each gut.
- For each gut:
 - Cut off the pieces of fat on the rod.
 - grab the end of the gut, drag it off the rod and into a straight line on the late.
 - Cut the cut through the middle, DO NOT CUT ALL THE WAY THROUGH THE OTHER END.
 - Grab the gut and dip it in the little beaker of pbs to clean it.
 - Lay the gut back onto the plate.
 - Lay the gut into the corresponding elongated well and make sure its straight.
- Wait 1 hour (Probably dont need to wait tbh since it'll take a long time for all the guts)
- For each gut:
 - Grab gut and put it on plate straight.
 - Find the notch that hasn't been cut yet.
 - Open the gut up so that the inside faces up on the plate. use the knitting needle rod things to roll the gut open on the plate.
 - You can now cut the bad end of the gut off.
 - Grab one of the pink base needles with the wax on the end.
 - ROLL FROM THE BOTTOM SO THE VILLI ARE FACING OUT
 - Once rolled, Take the gut off the wax needle with forceps.
 - Hold ends of the rolled gut with forceps and stick a 30 gage pin all the way through.
 - Place into the 6wp in the correct well.
- Wrap 6wp in aluminum foil and put into cold room 4C overnight.
- Once done, put 6wp into cold room

NEXT DAY

- Pour PFA into waste container.
- Fill wells with PBS to wash. Wait 10 min. Pour into waste.
- Wash livers with PBS for 10 min as well.
- Transfer into 15% sucrose until tissue sinks.